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CalibrationPerformanceMeasure.pdf	Main Document - PDF	2.1 MB	Page 4

Lumped-Parameter Modeling, Calibration, and Performance Measure of Flux Generation in Multipole Electromagnetic Actuators

Chia-Hsiang Menq, Chien-Chi Kuo, Hong-Yi Fan, Sheng-Shian Li and Fei Long

***Abstract*— This article presents a framework for lumped-parameter modeling, calibration, and quantitative performance evaluation of magnetic-flux generation in multipole electromagnetic actuators and magnetic tweezers. Utilizing a multiple-charge representation, the framework enables both input-current-based and sensor-voltage-based lumped-parameter modeling to replace computationally expensive Finite Element Method (FEM) simulations during the design phase. A five-phase calibration procedure combining nonlinear minimization and analytical calculations is proposed, enabling the identification of 18 geometric parameters, a global actuation gain, and a normalized flux-distribution matrix for hexapole systems using only 13 calibration points. The framework is validated against FEM simulations of a hexapole magnetic-tweezers design equipped with integrated Hall sensors, achieving a mean absolute error percentage (MAEP) of 0.35% across 1,771 finite element nodes within a 150 μm radius workspace. Building on the calibrated model, position-dependent metrics are formulated to quantify flux-generation efficiency and magnetic-flux manipulability. These metrics enable direct, standardized comparisons across alternative actuator designs during the design stage and establish a unified quantitative foundation for evaluating and benchmarking multipole electromagnetic actuators and magnetic-tweezers systems.**

***Index Terms* — Multipole electromagnetic tweezers, magnetic flux generation, lumped-parameter modeling, calibration, quantitative performance measure.**

I. Introduction

Multipole electromagnetic actuators (EMAs) and multipole magnetic tweezers (MTs) both employ multiple electromagnetic poles to generate and shape magnetic fields within a designated workspace. However, they are intended for fundamentally different application domains and therefore differ substantially in operating scale, control objectives, and precision requirements.

EMAs function like high-precision steering systems, designed to generate controlled kinetic motion on mesoscopic or microscopic targets. Depending on the application, targets can be multi-axis scanning probes [1,2], nickel-coated microrobots [3-5], miniatured swimming machines [6-9], mechanical platforms [10,11], medical devices [12-14], or micro-lenses [15]. By dynamically changing the current on multiple electro-magnetic poles (e.g., quadrupole or hexapole configurations), EMAs can steer, hover, and propel objects within a two- or three-dimensional workspace [16-18]. These systems employ active control and over-actuation.

Because magnetic field strength decays rapidly with distance, complex closed-loop feedback algorithms (typically using Hall sensors embedded in the tips of magnetic poles) are required to continuously stabilize and map the current to prevent the target from snapping to the magnetic poles. The required forces typically range from millinewtons (mN) to the microscale to overcome fluid drag, gravity, or mechanical stiffness, while the displacement ranges from micrometers to centimeters, e.g., guiding a microrobot through dense cellular environments or navigating in vascular models.

In contrast, MTs operate at a much finer biophysical scale, designed to probe biological processes closely associated with interactions ranging from subpiconewtons (pN) to tens of piconewtons [19-21], such as manipulating macromolecules like DNA, RNA, and proteins [22-24], probing cell membranes [25,26], and measuring intracellular properties [27]. These conventional passive force clamps can be used as simple force applicators or transducers. In a typical setup, one end of the molecule of interest is anchored to a targeted surface, and the other end is fixed to a superparamagnetic microbead.

In most setups, pairs of pointed electromagnetic poles are symmetrically arranged to concentrate magnetic flux into the three-dimensional workspace enclosed by the pole tips. They are used to apply magnetic fields to magnetize magnetic microbeads and exert forces or torques to achieve the desired pulling, twisting, and holding forces [28]. The displacement ranges from nanometers to micrometers (the length of a stretched DNA strands or proteins). Feedback mainly relies on high-speed video microscopy to measure the displacement of magnetic beads fixed at the moving end, enabling quasi-static measurements of molecular elasticity, folding and unfolding dynamics, and enzymatic reactions [29-31]. Although these setups are popular biophysical tools, their applications are limited to simple experiments on isolated biological samples.

Recently, hexapole electromagnetic actuators equipped with six integrated Hall sensors and employing advanced control techniques have been developed as actively controlled magnetic tweezers for probing living cells at delicate biophysical scales [32-35]. These systems utilize six independently driven pointed electromagnetic poles to generate and shape magnetic fields within a three-dimensional workspace enclosed by the pole tips [36-37]. Combined with advanced visual servo control

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technology [18], they can manipulate a single untethered magnetic scanning microprobe in aqueous solution at high speed under a microscope, which is quite different from conventional magnetic tweezers where the probe is confined to a surface [29-31]. From a system perspective, these actuators are six-input distributed-parameter nonlinear systems, which can be represented as six-input current-driven systems with position-dependent three-output magnetic flux response.

Although advanced control can achieve stable operation of such complex six-input-driven systems, the performance, such as efficiency, manipulability, and controllability, largely depends on their design. While finite element method (FEM) simulations can be used to model complex flux generation processes during the design phase, lumped-parameter models are more suitable at this stage for quantitatively evaluating actuator performance and comparing different designs.

Therefore, this paper proposes a general framework for lumped parameter modeling, calibration, and quantitative performance evaluation of flux generation processes in multipole electromagnetic actuators. Although this framework is applicable to any multipole configuration, a hexapole actuator design with integrated Hall sensors is used throughout the paper as a representative example. The remainder of the paper is organized around three major contributions, as described below.

Section II presents a current-based lumped-parameter model formulated using a multiple-charge representation, in which an actuation gain and a normalized flux-distribution matrix are used to model magnetic-flux generation and coupling from six sources. Furthermore, for the hexapole system, a 6×6 dimensionless spatial-distribution matrix parameterized by 18 geometric parameters is employed to capture the spatial flux distribution within the workspace. A five-phase calibration procedure is proposed to identify a complete lumped-parameter model, which can be used to predict flux generation and spatial distribution within the workspace and evaluate the flux-generation performance of the hexapole system design. Finite-element method (FEM) simulations are then used to validate both the modeling approach and calibration method.

Section III develops, calibrates, and validates a sensor-voltage-based lumped-parameter model. By using sensor voltage instead of coil current, this model reduces the uncertainty caused by remanence and hysteresis in current-based modeling. The influence of sensor placement is also investigated. Based on the calibrated model, Section IV introduces position-dependent metrics for quantitatively evaluating flux-generation performance. These metrics enable direct, standardized comparisons of different actuator designs during the design phase.

Section V concludes the paper. The proposed calibration method is applicable to both actuator design based on the finite element method and experimental physical actuators, and the proposed position-dependent metrics provide a unified basis for evaluating and comparing multipole electromagnetic actuators and magnetic tweezers.

II. Current-Based Lumped-Parameter Modeling and Calibration

This section discusses lumped-parameter modeling and parameter calibration. Finite-element method (FEM) simulations together with the calibrated lumped-parameter model validate both the modeling approach and the proposed calibration procedure.

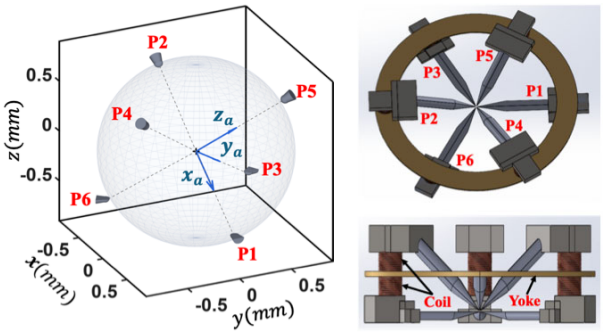


Fig.1. Hexapole electromagnetic actuator design: (L) Three pairs of electromagnetic pole tips are arranged symmetrically along the three axes of the actuator coordinate system to enclose the 3D workspace, (R) The CAD model of a benchmark design in which six sharp-tipped poles are integrated through a magnetic yoke to enhance the generation of magnetic flux and each pole is driven by an independent coil.

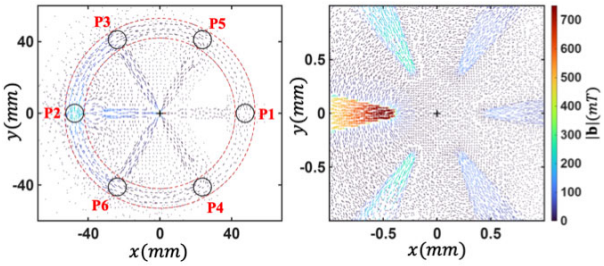


Fig.2. Results of a finite element method (FEM) simulation when the 70-turn coil of P2 is excited with a current of 1 A: (L) A top view of the magnetic flux path throughout the entire hexapole system, (R) Enlarged view of the spatial distribution of magnetic flux generated near the six pole tips and within the workspace.

A. Multiple-electromagnetic-pole actuation

In most multi-pole electromagnetic actuators or tweezers, pairs of pointed electromagnetic poles, i.e., n_p poles in total, are symmetrically arranged to concentrate magnetic flux into a n -dimensional workspace enclosed by the pole tips, assuming that $n_p \geq 2n$. In the following discussion, we assume $n_p = 6$, i.e., a hexapole actuator, and $n = 3$, i.e., a three-dimensional workspace.

Fig.1 illustrates the design of a hexapole electromagnetic actuator or tweezers: (L) It employs three pairs of pointed electromagnetic poles. Their pole tips are arranged symmetrically along the three axes of the actuator's coordinate system, concentrating the magnetic flux into the three-dimensional space enclosed by the six pole tips, within which specimen samples and the micro probe reside. (R) The CAD model of a benchmark design. It shows that each pole driven by the current of an independent coil. Furthermore, they are connected by a ferromagnetic yoke to form a closed magnetic circuit, thereby enhancing the generation of magnetic flux.

ANSYS Mechanical APDL is used to perform FEM simulations. The magnetic flux path throughout the entire hexapole system under unipolar excitation is shown in the left panel of Fig.2. The spatial distribution of magnetic flux near the six pole tips and within the workspace is shown in the right panel of

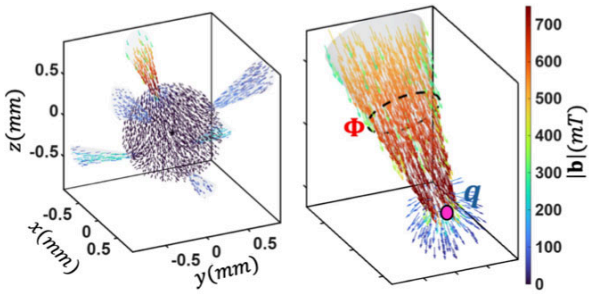


Fig.3. (L) The magnetic flux flow in the pole tips and within the spherical workspace, (R) The generated magnetic flux is confined in P2 and radiates from the pole tip.

Fig.2. By using a small thread hold ($0.1\,mT$) to filter out relatively small magnetic flux in the spatial distribution, the left panel of Fig.3 focusses on the magnetic flux flow in the pole tips and within the spherical workspace. It shows that most of the generated magnetic flux is confined within P2, diverging into and filling the spherical workspace, and flowing into other magnetic poles.

The right panel of Fig.3 shows that as the cross-sectional area of P2 decreases, the magnetic flux becomes more concentrated at the pole tips. In regions with sufficiently small cross-sectional areas, the magnetic flux begins to radiate from the surface near the pole tip.

B. Multi-charge model

When observing the individual poles of the actuator from within the workspace, the magnetic flux is concentrated near the pole tips and radiates outward from each pole tip. The magnetic flux appears as if it originates from independent point sources. Therefore, the magnetic monopole approximation [38] can be used and extended to a magnetic hexapole model to depict the spatial distribution of the magnetic field within the three-dimensional workspace. In this magnetic hexapole model, the magnetic field generated by each pole is approximated as the magnetic field of a point charge, e.g., the charge q shown in the right panel of Fig.3, while the total magnetic field generated by the actuator is obtained through the superposition principle.

As the magnetic flux Φ is permeated through the magnetic pole to the pole tip when current being applied to its actuating coil, a point magnetic charge associated with the magnetic flux (Fig.3 (R)) can be defined in analogy to the electric charge as $q = \Phi/\mu_o$, where μ_o is the magnetic permeability of vacuum [39]. The resulting position-dependent magnetic flux density produced by the charge in the workspace can, therefore, be written as $\mathbf{b}(\mathbf{p}) = (\mu_o/4\pi)[q \cdot (\mathbf{p} - \mathbf{p}_c)/\|\mathbf{p} - \mathbf{p}_c\|^3]$, where $\mathbf{p}$ is any spatial point in the workspace, $\mathbf{p}_c$ is the position of the charge.

By introducing a characteristic length scale ℓ to normalize position vectors, i.e., $\bar{\mathbf{p}} = \mathbf{p}/\ell$ and $\bar{\mathbf{p}}_c = \mathbf{p}_c/\ell$, a dimensionless 3×1 spatial distribution vector is defined to depict the spatial distribution of the magnetic flux attributed to the specific pole,

$$\mathbf{s}_c(\bar{\mathbf{p}}) = \frac{\bar{\mathbf{p}} - \bar{\mathbf{p}}_c}{\|\bar{\mathbf{p}} - \bar{\mathbf{p}}_c\|^3}. \tag{1}$$

Accordingly, the resulting position-dependent magnetic flux density produced by a hexapole actuator in the three-dimensional workspace can be written as

$$\mathbf{b}(\mathbf{p}, \mathbf{Q}; \ell) = \left(\frac{\mu_o}{4\pi}\right)\frac{1}{\ell^2}\mathbf{S}(\bar{\mathbf{p}})\mathbf{Q}, \tag{2}$$

where $\mathbf{Q} = [q_1 \quad q_2 \quad q_3 \quad q_4 \quad q_5 \quad q_6]^T$ is the 6×1 magnetic-charge vector and $\mathbf{S}(\bar{\mathbf{p}})$ is the 3×6 spatial-distribution matrix defined as follows,

$$\mathbf{S}(\bar{\mathbf{p}}) = \left[\underbrace{\frac{\bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_1}}{\|\bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_1}\|^3}}_{\mathbf{s}_{c1}} \quad \underbrace{\frac{\bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_2}}{\|\bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_2}\|^3}}_{\mathbf{s}_{c2}} \quad \cdots \quad \underbrace{\frac{\bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_6}}{\|\bar{\mathbf{p}} - \bar{\mathbf{p}}_{c_6}\|^3}}_{\mathbf{s}_{c6}}\right]. \tag{3}$$

Therefore, using a 6-charge model, a six-input distributed parameter nonlinear actuator can be represented as a six-charge lumped parameter system with a position-dependent output flux response in a three-dimensional workspace. This approximation yields an expression which is very similar to that of an electric field. More importantly, it simplifies the representation of the spatial distribution of the magnetic field in the workspace [40].

Since the actual six inputs to the actuator are six coil currents, in a current-based magnetic-flux model, a relationship between the 6×1 input current vector $\mathbf{I}$ and the 6×1 magnetic flux vector $\mathbf{\Phi}$ needs to be established. When ignoring the effects of remanence, hysteresis, and magnetization dynamics, a quasi-static relationship can be established through magnetic circuit analysis or finite element analysis.

The magnetic flux is produced by the magnetomotive force, which is proportional to the input current, $\mathbf{I}$, and the turns of the coil, N_T. By Hopkinson's law in magnetic circuit analysis, the magnetic flux is quasi-statically related to the input current by $\mathbf{\Phi} = (N_T/\mathcal{R}_a)\mathbf{KI}$, where $\mathcal{R}_a$ is the lumped magnetic reluctance and $\mathbf{K}$ is the flux-distribution matrix. The theoretical value of the 6×6 flux-distribution matrix has 5/6 as the diagonal entries and -1/6 as the off-diagonal entries [37]. It was, however, shown in [36] that due to the magnetic leakage and the discrepancy in geometry improved estimation of $\mathcal{R}_a$ and $\mathbf{K}$ can be obtained using FEM simulation and calibration. Nonetheless, a current-based-magnetic-flux model can be derived,

$$\mathbf{b}(\mathbf{p}, \mathbf{I}; \ell) = \left(\frac{N_T}{\mathcal{R}_a}\right)\left(\frac{1}{4\pi}\right)\frac{1}{\ell^2}\mathbf{S}(\bar{\mathbf{p}})\mathbf{KI}, \tag{4}$$

C. Charge-location model

Although the six pole tips are placed symmetrically along the three axes of the actuator coordinate system in the design, the results in Figure 3 show that the effective point sources are clearly offset from the design tip. Furthermore, due to geometric constrains and kinematic requirement, the body axis of each individual pole may not be perfectly aligned with its corresponding coordinate axis. For example, the three conical magnetic lower poles in the design shown in Fig.1 (R) are milled in half to form a plane and are leveled with the horizontal plate of the microscope to support the coverslip being inserted into the workspace. Therefore, whereas these six-point sources locate near the intersections of the three coordinate axes and a sphere with a radius greater than the design value (i.e., $\ell > \ell_o$), they are likely not at these intersections themselves.

Specifying the six point-source locations completely requires 18 geometric parameters. Instead of using six 3×1 position vectors, together with the radius ℓ, we introduce a 17-

offset-parameter representation of the normalized charge positions to define the six positions, namely:

$$[\overline{\mathbf{p}}_{c1} \quad \overline{\mathbf{p}}_{c2} \quad \overline{\mathbf{p}}_{c3} \quad \overline{\mathbf{p}}_{c4} \quad \overline{\mathbf{p}}_{c5} \quad \overline{\mathbf{p}}_{c6}] = \mathbf{U} + \mathbf{E}, \tag{5}$$

where $\mathbf{U}$ is constructed using three standard basis unit vectors, i.e., $\mathbf{U} = [\hat{\mathbf{i}} \quad -\hat{\mathbf{i}} \quad \hat{\mathbf{j}} \quad -\hat{\mathbf{j}} \quad \hat{\mathbf{k}} \quad -\hat{\mathbf{k}}]$, and $\mathbf{E}$ is a dimensionless 3×6 offset matrix, in which each column represents the normalized offset vector corresponding to the magnetic-charge position. The z-component e_{6z} of the *sixth* offset vector is not an independent parameter, but is related to five other parameters,

$$e_{6z} = e_{1x} - e_{2x} + e_{3y} - e_{4y} + e_{5z}. \tag{6}$$

By stacking all 17 offset parameters into a single 17×1 vector $\mathbf{e}$ and normalizing the first diagonal element of the flux-distribution matrix $\mathbf{K}$ to its theoretical value 5/6, i.e., $\overline{\mathbf{K}} = 5/(6|k_{11}|)\,\mathbf{K}$, the current-based-magnetic-flux model can be expressed as follows:

$$\mathbf{b}(\mathbf{p}, \mathbf{I}; \ell, \mathbf{e}) = \underbrace{\left(\frac{N_T}{\mathcal{R}_a}\right)\left(\frac{1}{4\pi}\right)\frac{1}{\ell^2}\left(\frac{6|k_{11}|}{5}\right)}_{{}^{B}g_I}\mathbf{S}(\mathbf{p}/\ell, \mathbf{e})\overline{\mathbf{K}}\mathbf{I}, \tag{7}$$

where ${}^{B}g_I$ is the actuation-gain of the actuator. Together with the 18 parameters, the spatial distribution matrix $\mathbf{S}(\mathbf{p}/\ell, \mathbf{e})$ characterizes the spatial distribution of the generated magnetic flux in the workspace. When assuming the 17×1 vector $\mathbf{e}$ to be zero, a simplified single parameter model $\mathbf{S}(\mathbf{p}/\ell)$ might be employed to render acceptable modeling accuracy, depending on the design configuration of the actuator and the accuracy requirement.

D. Calibration

Let ${}^{B}\mathbf{H}_I = {}^{B}g_I\overline{\mathbf{K}}$, it follows directly from (7) that the accuracy of the model depends on accurate knowledge of the 6×6 matrix ${}^{B}\mathbf{H}_I$ (36 parameters), as well as the geometric parameters ℓ and $\mathbf{e}$ (18 parameters) that enter the spatial-distribution matrix $\mathbf{S}(\mathbf{p}/\ell, \mathbf{e})$. To obtain these parameters, we propose a five-phase calibration procedure, as described below.

Phase 1 – Data Acquisition

1. Selection of spatial calibration points

   Choose N_c points $\mathbf{p}_i \in R^3$ for $i = 1,2,\cdots,N_c$ within the three-dimensional workspace.

2. Excitation design

   Apply N_I distinct six-input current excitations using current vectors $\mathbf{I}_j$ for $j = 1,2,\cdots,N_I$. Construct the combined excitation-matrix $\mathbf{F} = [\mathbf{I}_1 \quad \mathbf{I}_2 \quad \cdots \quad \mathbf{I}_{N_I}]$, and ensure that $rank(\mathbf{F}) = 6$ so that all six poles are sufficiently excited.

3. Magnetic-flux-density measurements

   For each excitation $\mathbf{I}_j$, record the magnetic flux density at each spatial point: $\mathbf{b}_{ij} = \mathbf{b}(\mathbf{p}_i, \mathbf{I}_j) \in R^3$, $i = 1,2,\cdots,N_c$ and $j = 1,2,\cdots,N_I$.

Phase 2 – Model Prediction

For each measurement pair $(\mathbf{p}_i, \mathbf{I}_j)$, the model predicts

$$\mathbf{b}_{ij} \approx \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}})\,{}^{B}\hat{\mathbf{H}}_I\mathbf{I}_j, \tag{8}$$

where $\hat{\ell}$, $\hat{\mathbf{e}}$, and ${}^{B}\hat{\mathbf{H}}_I$ are the unknown parameters to be calibrated.

Phase 3 – Geometric-Parameter-Cost-Function Construction

1. Define the spatial-distribution weighting matrix:

$$\mathbf{W}(\hat{\ell}, \hat{\mathbf{e}}) = \sum_{i=1}^{N_c}\mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}})\mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}). \tag{9}$$

2. Project each measured flux density onto the spatial-distribution basis:

$$\tilde{\mathbf{b}}_{ij}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) = \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}})\mathbf{b}_{ij}. \tag{10}$$

3. The overall cost function is:

$$J(\hat{\ell}, \hat{\mathbf{e}}) = \sum_{j=1}^{N_I}\left\{\sum_{i=1}^{N_c}\tilde{\mathbf{b}}_{ij}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}})\left(\mathbf{W}(\hat{\ell}, \hat{\mathbf{e}})\right)^{-1}\sum_{i=1}^{N_c}\tilde{\mathbf{b}}_{ij}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}})\right\}. \tag{11}$$

Phase 4 – Cost Function Minimization

Solve the optimization problem: $Minimize\ J(\hat{\ell}, \hat{\mathbf{e}})$ to obtain the 18 calibrated geometric parameters $\hat{\ell}$ and $\hat{\mathbf{e}}$.

Phase 5 – Calculation of ${}^{B}g_I$ and $\overline{\mathbf{K}}$

1. For each excitation $\mathbf{I}_j$, compute

$$\boldsymbol{\eta}_j = \left(\mathbf{W}(\hat{\ell}, \hat{\mathbf{e}})\right)^{-1}\sum_{i=1}^{N_c}\tilde{\mathbf{b}}_{ij}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \tag{12}$$

2. Form the $6 \times N_I$ matrix:

$$\boldsymbol{\Theta} = [\boldsymbol{\eta}_1 \quad \boldsymbol{\eta}_2 \quad \cdots \quad \boldsymbol{\eta}_{N_I}] \tag{13}$$

3. Calculate the input-current-flux matrix:

$${}^{B}\hat{\mathbf{H}}_I = \boldsymbol{\Theta}\mathbf{F}^T(\mathbf{F}\mathbf{F}^T)^{-1} \tag{14}$$

4. Extract the actuation gain and the normalized megnetic-flux-distribution matrix:

$${}^{B}\hat{g}_I = \frac{6}{5}\left|\hat{h}_{11}\right| \text{ and } \hat{\overline{\mathbf{K}}} = \frac{5}{6\hat{h}_{11}}\,{}^{B}\hat{\mathbf{H}}_I \tag{15}$$

It is worth noting that once the 18 geometric parameters are obtained through nonlinear minimization in the proposed

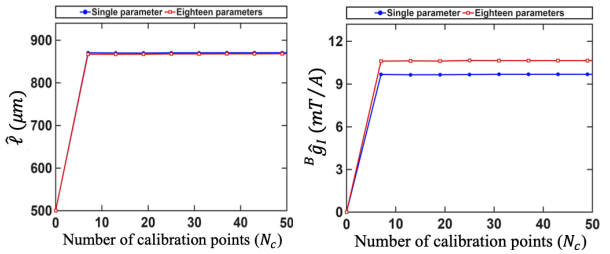


Fig.4. Relationship between the number of calibration points and the estimated parameter values.

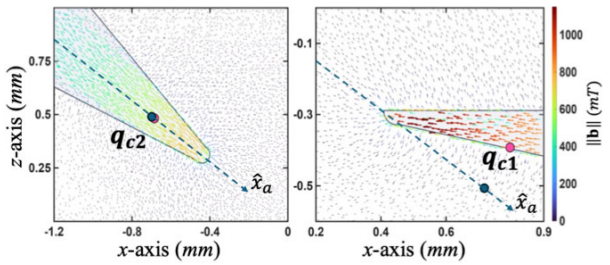


Fig.5. Comparison of calibrated charge positions using different numbers of geometric parameters, i.e., using 18 parameters (red) versus using only a single parameter (blue): (left) calibrated charge positions of P2 (upper pole) and (right) calibrated charge positions of P1 (lower pole).

calibration procedure, the actuation gain and the normalized flux-distribution matrix can be obtained analytically.

E. Validation

Six finite element simulations were performed using ANSYS Mechanical APDL, with six specific input-current vectors selected such that the combined excitation matrix $\mathbf{F}$ is the 6×6 identity matrix $\mathbf{I}_{6\times6}$. The three-dimensional workspace was a sphere of a specified radius. Following the proposed calibration procedure, N_c spatial points were uniformly distributed along the three actuation axes within the specified workspace and the magnetic flux density at these points was used for calibration. Notably, due to the use of the results from six finite element simulations, the amount of data ($\mathbf{b}_{ij}$) used for calibration is $6N_c$.

The following calibration results aim to validate the proposed calibration framework, confirm the number of points required for reliable calibration, and illustrate the importance of using 17 offset parameters in the calibration model. The first calibration result was obtained using the magnetic flux density within a workspace with a radius of 150 μm. It is worth noting that although there are 1,771 finite element nodes within the specified workspace, the number of spatial points used for calibrating the geometric parameters is $N_c = 1 + 6n$, where $n = 1, 2, \cdots, 8$. This paper simultaneously performs calibration using a single-geometric-parameter calibration model and the proposed 18-geometric-parameter calibration model and compares the results.

Figure 4 shows the relationship between the number of calibration points and the estimated parameter values. It clearly demonstrates that as the number of calibration points increases, both the estimated $\hat{\ell}$ and ${}^B\hat{g}_I$ converge to their respective final values. Furthermore, although the final value of $\hat{\ell}$, 868.57 μm, differs significantly from the design value (500 μm), the estimated $\hat{\ell} = 866.84\ \mu m$ is close to its final value when using seven points symmetrically distributed along the three axes. More specifically, when $N_c = 13$ and 18 geometric parameters are used in the calibration model, the obtained $\hat{\ell} = 867.20\ \mu m$ deviates from its final values by less than 0.2 %. Using the calibrated 17-parameter vector $\hat{\mathbf{e}}$, the dimensionless offset-matrix is obtained,

$$\hat{\mathbf{E}} = \begin{bmatrix} 0.014 & 0.021 & -0.117 & 0.001 & -0.000 & -0.118 \\ -0.114 & 0.002 & 0.020 & 0.019 & 0.000 & -0.116 \\ 0.115 & -0.003 & 0.116 & -0.001 & -0.011 & -0.018 \end{bmatrix}. \quad (16)$$

It clearly shows that the offsets associated with the three upper poles, i.e., P2, P4, and P5, are an order of magnitude smaller than the offsets associated with the three lower poles.

Figure 5 compares the estimated charge positions of P1 (lower pole) and P2 (upper pole) when calibrating the model using 18 geometric parameters with those using only a single geometric parameter. In the design, relative to the microscope coordinate system, the tip of P2 is placed at $[-0.4 \quad 0 \quad 0.3]^T$, and the tip of P1 is located at $[0.4 \quad 0 \quad -0.3]^T$. Since the conical body of P2 is aligned with the $\hat{x}_a$-axis of the actuator coordinate system, the estimated charge position (blue dot) obtained from single-parameter calibration remains on the $\hat{x}_a$-axis, differing from the calibrated position (red dot) obtained from 18-parameter calibration by only about 18 μm. On the other hand, P1 is one of the lower poles, which are milled in half and leveled with the horizontal plate of the microscope. Although the tip of P1 is placed on the $\hat{x}_a$-axis, the body axis of P1 is not aligned with the $\hat{x}_a$-axis. When using single-parameter calibration, the calibrated charge location (blue dot) must remain on the $\hat{x}_a$-axis, and therefore it deviates significantly from the more accurate location (red dot) calibrated through the 18-parameter calibration.

Using the 18 obtained geometric parameters, the actuation gain and the normalized flux-distribution matrix can be calculated. The calculation results show that the deviation between ${}^B\hat{g}_I = 10.62\ mT/A$ and its final values is less than 0.2 %, and the flux-distribution matrix is nearly diagonally dominant, i.e.,

$$\hat{\mathbf{K}} = \begin{bmatrix} 0.833 & -0.095 & -0.200 & -0.166 & -0.173 & -0.197 \\ -0.215 & 0.963 & -0.218 & -0.151 & -0.157 & -0.214 \\ -0.191 & -0.167 & 0.826 & -0.097 & -0.173 & -0.197 \\ -0.207 & -0.151 & -0.224 & 0.958 & -0.156 & -0.213 \\ -0.208 & -0.151 & -0.218 & -0.151 & 0.956 & -0.221 \\ -0.190 & -0.167 & -0.200 & -0.167 & -0.102 & 0.828 \end{bmatrix}. \quad (17)$$

The flux-distribution matrix shows that when current is applied to its coil, the generated magnetic flux is confined within the specific pole and permeated through the magnetic pole to its tip, resulting in significantly larger diagonal elements in the flux-distribution matrix. The magnetic flux then diverges into and fills the workspace, and flows into other magnetic poles, yielding smaller and negative off-diagonal elements in the flux-distribution matrix. Notably, these six diagonal elements are divided into two groups. The group with larger values corresponds to the three upper poles, namely P2, P4, and P5, whereas the group with smaller values corresponds to the three lower

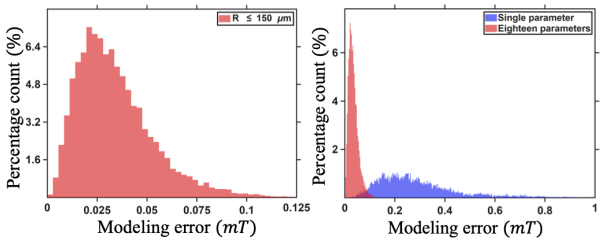


Fig.6. Modeling errors produced by using the calibrated models obtained using two calibration methods.

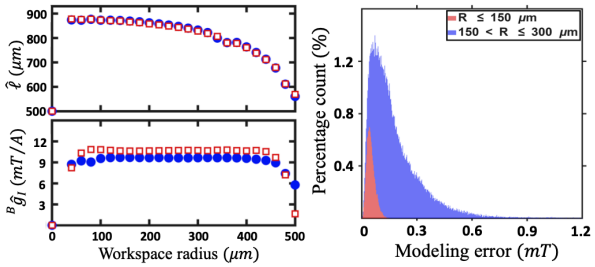


Fig.7. (L) Relationship between estimated characteristic length and actuation gain and the specified workspace radius: (Red) 18-parameter model and (Blue) single-parameter model. (R) Histogram of modeling errors at 14082 nodes within a specified workspace with a radius of 300 μm.

poles as they are milled in half and leveled with the horizontal plate of the microscope.

Figure 6 compares the resulting modeling errors produced by using the calibration models obtained using the two calibration methods. Notably, the single-parameter calibration method uses 25 points, while the 18-parameter calibration method uses 13 points. The left panel of Figure 6 shows the histogram of modeling errors evaluated at the 1771 finite element nodes within the specified workspace for the model obtained using the 18-parameter calibration method. The mean of modeling errors is 0.0340 mT, while the average magnetic field strength is 10.2452 mT. The right panel compares the histograms of modeling errors for the models obtained using the two calibration methods. The results show that single-parameter calibration leads to a larger modeling error, attributed to an increase in charge location calibration error. The mean of modeling errors increases from 0.0340 mT to 0.2878 mT.

Figure 7 (left) illustrates the relationship between the calibrated parameters (i.e., characteristic length $\hat{\ell}$ and actuation gain ${}^{B}\hat{g}_I$) obtained using two calibration methods and the specified workspace radius. The results clearly show that the characteristic lengths obtained by the two calibration methods accord well. On the other hand, the results indicate that when the specified workspace radius is small, the reliability of the calibrated actuation gain is low due to insufficient spatial variation of the magnetic field generated within the relatively small spherical workspace. When the radius exceeds approximately 120 μm, the calibrated actuation gain remains constant until the radius reaches 400 μm, which is 80 % of the design value.

As described in Section II-B, the multi-charge model is only valid when the individual poles of the magnetic tweezers are observed from a distance, where the magnetic flux appears as if it originates from independent point sources. Therefore, when the radius of the specified workspace is too large, e.g., exceeding 400 μm, the reliability of the obtained calibration model

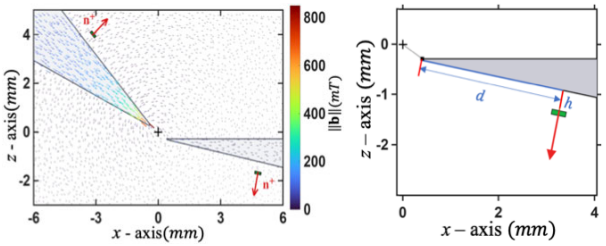


Fig.8. Hall sensors are integrated into a hexapole system to function as a distributed sensing network, where each sensor is mounted on a surface near the tip of a pole.

decreases, and the resulting modeling error increases dramatically for spatial points close to any pole tip.

Figure 7 (right) shows a histogram evaluating the modeling error of the model obtained using the 18-parameter calibration method on 14082 finite element nodes within a specified workspace with a radius of 300 μm for the model. These 14082 finite element nodes are divided into two groups. The first group contains 1771 nodes located within a sphere with a radius of 150 μm, and their corresponding modeling errors are shown in red. The remaining nodes belong to the second group, and their corresponding modeling errors are shown in blue. The results clearly show that the larger the radial position of the spatial point, the greater the modeling error.

III. Hall-sensor-voltage-based lumped-parameter modeling and calibration

While the actuator is driven by six coil currents, the magnetic field in the workspace is ultimately shaped by the magnetic flux concentrated at the six pole tips. Hysteresis introduces notable uncertainties during current actuation, resulting in substantial modeling errors [41-44]. Furthermore, extending the previously developed method [45] to compensate for coupled hysteresis across multiple poles in a hexapole actuator is highly complex. Consequently, directly measuring the magnetic flux using Hall sensors presents a far more effective and practical solution.

A. Voltage-based model

Although the concentrated magnetic flux at each pole tip cannot be directly measured, the hexapole system integrates six Hall sensors to form a distributed sensing network. Figure 8 (left) illustrates two Hall sensors integrated into the first pair of poles (P1 and P2) of the tweezers. Each sensor is mounted on a surface close to the pole tip, producing an output voltage proportional to the local magnetic flux passing through it. It primarily measures the magnetic flux at its own pole tip while receiving smaller crosstalk fields from the other five pole tips. Figure 8 (right) illustrates the placement of each sensor. The sensors are placed at a distance d from the pole tip and are closely fitted to the selected surface, such that the detector is parallel to the surface, with the gap h determined by the thickness of the sensor housing. The outputs of the six sensors collectively encode the flux distribution of the entire workspace of the hexapole system.

By superposition, the resulting 6×1 voltage vector $\mathbf{v}$ is a linear transformation of the 6×1 pole-tip flux vector $\mathbf{\Phi}$.

Conversely, applying the inverse transformation to $\mathbf{v}$ yields the flux measurements at all six pole tips,

$$\mathbf{\Phi} = \mathbf{M}\mathbf{v}, \tag{16}$$

where $\mathbf{M}$ is the sensor-network measurement matrix. Due to crosstalk between pole tips and leakage of the coil magnetic field, this 6×6 measurement matrix is not diagonal, contrary to the assumption in [36]. If each sensor has strong locality (depending on sensor placement and coil position), this matrix may be diagonally dominant. In the proposed calibration method, the sensor-network measurement matrix is identified as a complete 6×6 matrix, thus making the calibration results robust to sensor placement.

Since $\boldsymbol{\phi} = (N_T/\mathcal{R}_a)\mathbf{K}\mathbf{I} = \mathbf{M}\mathbf{v}$, using (7) and normalizing the first diagonal element of $\mathbf{M}$ so that the corresponding flux-voltage gain matches the specified value 5/6 in $\bar{\mathbf{M}}$, i.e., defining $\bar{\mathbf{M}} = 5/(6m_{11})\,\mathbf{M}$, the Hall-sensor-voltage-based flux model can be expressed as follows:

$$\mathbf{b}(\mathbf{p}, \mathbf{v}; \ell, \mathbf{e}) = \underbrace{\left(\frac{1}{4\pi}\right)\frac{1}{\ell^2}\left(\frac{6|m_{11}|}{5}\right)}_{{}^{B}g_V}\mathbf{S}(\mathbf{p}/\ell\,, \mathbf{e})\bar{\mathbf{M}}\mathbf{v}, \tag{17}$$

where ${}^{B}g_V$ is proportional to the inverse of the measurement gain of the sensor network.

B. Calibration

Let ${}^{B}\mathbf{H}_V = {}^{B}g_V\bar{\mathbf{M}}$, the same five-phase calibration procedure proposed in Section II-D with one additional measurement and two revisions is proposed to calibrate the spatial parameters ℓ and $\mathbf{e}$ (18 parameters) and subsequently calculate the 6×6 sensor-voltage-flux matrix ${}^{B}\mathbf{H}_V$. The addition and revisions are listed below.

Addition – Hall-sensor-voltage measurements

For each excitation $\mathbf{I}_j$, record $\mathbf{v}_j \in R^6$, $j = 1,2,\cdots,N_I$ and construct the combined voltage matrix $\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_{N_I}]$.

Revision #1 – Model Prediction

For each pair of measurements $(\mathbf{p}_i, \mathbf{v}_j)$, the model predicts

$$\mathbf{b}_{ij} \approx \mathbf{S}\big(\mathbf{p}_i/\hat{\ell}\,, \hat{\mathbf{e}}\big)\,{}^{B}\hat{\mathbf{H}}_V\mathbf{v}_j, \tag{18}$$

where $\hat{\ell}$ and $\hat{\mathbf{e}}$ are 18 unknown parameters that need to be calibrated first in the third and fourth phases of the proposed calibration method.

Revision #2 – Calculation of ${}^{B}g_V$ and $\bar{\mathbf{M}}$

Calculate the sensor-voltage-flux matrix:

$${}^{B}\hat{\mathbf{H}}_V = \mathbf{\Theta}\mathbf{V}^T(\mathbf{V}\mathbf{V}^T)^{-1} \tag{19}$$

Extract the actuation gain and the normalized sensor-network measurement matrix:

$${}^{B}\hat{g}_V = \frac{6}{5}|\hat{m}_{11}| \text{ and } \hat{\bar{\mathbf{M}}} = \frac{5}{6\hat{m}_{11}}\,{}^{B}\hat{\mathbf{H}}_V \tag{20}$$

C. Validation

The same six simulation experiments described in Section II-E are used to validate the proposed calibration method for the sensor-voltage-based model. The required Hall sensor voltage measurement is performed in two steps: first, six Hall sensors are placed, and then the sensor voltage readings are acquired. The position of each Hall sensor is represented by a spatial point, the position of which is determined by the design position of the corresponding magnetic pole, the selected surface, the planned distance d, and the gap h of the selected sensors. The voltage reading of the kth sensor in the jth simulation experiment is calculated as follows:

$$v_{kj} = s_H\hat{\mathbf{n}}_k \cdot \mathbf{b}_{kj}, \tag{21}$$

where s_H is the sensitivity of the Hall sensor used (mV/mT), $\hat{\mathbf{n}}_k$ is the surface normal of the kth sensor, and $\mathbf{b}_{kj} = \mathbf{b}_j(\mathbf{p}_k)$ is the magnetic flux density at the position of the kth sensor in the jth simulation experiment, which is determined by Barycentric interpolation [46].

This paper uses previously obtained geometric parameter calibration results to calibrate two sensor-voltage-based models with different sensor positions and compares the results. It should be noted that the calibration workspace radius is 150 μm, $N_c = 13$, and 18 geometric parameters are used in the calibration model. The obtained $\hat{\ell}$ value is 867.20 μm and the offset matric $\hat{\mathbf{E}}$ is given by equation (16).

Figure 8 illustrates the selection of sensor placement surfaces, with the upper surface used for upper poles and the lower surface used for lower poles. The gap h of selected sensors is 0.421 mm. In the first calibration, the sensor was placed at a distance d of 3 mm, and voltage readings were acquired,

$$\mathbf{V} = \begin{bmatrix} 1396 & -45 & -296 & -238 & -249 & -295 \\ -52 & 1599 & -236 & -508 & -515 & -238 \\ -294 & -234 & 1395 & -46 & -241 & -294 \\ -235 & -532 & -52 & 1584 & -528 & -226 \\ -242 & -521 & -230 & -520 & 1581 & -48 \\ -288 & -243 & -297 & -235 & -45 & 1383 \end{bmatrix}. \tag{22}$$

This matrix is used in formula (19) to calculate the 6×6 sensor-voltage-flux matrix ${}^{B}\mathbf{H}_V$, which is subsequently used in equation (20) to calculate ${}^{B}\hat{g}_V = 5.9476 \times 10^{-3}\,mT/mV$ and the normalized sensor-network measurement matrix,

$$\hat{\bar{\mathbf{M}}} = \begin{bmatrix} 0.833 & -0.451 & -0.191 & -0.397 & -0.401 & -0.190 \\ -0.012 & 1.171 & 0.003 & 0.354 & 0.356 & 0.003 \\ -0.208 & -0.437 & 0.832 & -0.474 & -0.434 & -0.208 \\ 0.042 & 0.409 & 0.018 & 1.227 & 0.411 & 0.030 \\ 0.037 & 0.402 & 0.025 & 0.403 & 1.244 & 0.005 \\ -0.201 & -0.414 & -0.197 & -0.412 & -0.463 & 0.846 \end{bmatrix}. \tag{23}$$

The results of equation (23) clearly show that the normalized sensor-network measurement matrix obtained using the proposed calibration method is far from a diagonal matrix, or even a diagonally dominant matrix. In fact, significant crosstalk between the poles and leakage of the coil magnetic field were observed in the finite element simulation results.

The mean absolute error percentage (MAEP) is introduced to measure the overall modeling error of the calibrated model in a specified workspace. It is the mean absolute difference between the modeled value, namely ${}^{B}\hat{g}_V\mathbf{S}(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}})\widehat{\mathbf{M}}\mathbf{v}_j$, and the corresponding true value ($\mathbf{b}_{ij}$), expressed as a percentage of the mean absolute true value,

$$\text{MAEP} = \frac{\sum_{j=1}^{6}\sum_{i=1}^{n}\left\|\mathbf{b}_{ij} - {}^{B}\hat{g}_V\mathbf{S}(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}})\widehat{\mathbf{M}}\mathbf{v}_j\right\|}{\sum_{j=1}^{6}\sum_{i=1}^{n}\|\mathbf{b}_{ij}\|}\cdot 100\%. \tag{24}$$

When using the available true values of all finite element nodes in the workspace with a radius of 150 μm, i.e., $n = 1771$, the quantitative measure of the modeling error of the calibrated model is MAEP = 0.35 %. However, when only a single geometric parameter is used in the calibration, the mean absolute error percentage increases to 2.81 %. Furthermore, if the normalized sensor-network measurement matrix is identified as a diagonal matrix, the mean absolute error percentage increases significantly even further.

In the second calibration, the sensor placement distance d was increased to 5 mm, and voltage readings were collected,

$$\mathbf{V} = \begin{bmatrix} 1159 & -24 & -214 & -209 & -202 & -218 \\ -26 & 1270 & -194 & -386 & -395 & -196 \\ -214 & -208 & 1163 & -23 & -202 & -213 \\ -195 & -391 & -26 & 1278 & -388 & -200 \\ -194 & -383 & -191 & -381 & 1259 & -25 \\ -215 & -206 & -219 & -213 & -23 & 1165 \end{bmatrix}. \tag{25}$$

The results show that the entry values of the second $\mathbf{V}$ matrix are reduced compared to the $\mathbf{V}$ matric in equation (22). At the same time, the calibration gain ${}^{B}\hat{g}_V = 7.5570\times10^{-3}\, mT/mV$ increases and the normalized sensor-network measurement matrix is significantly different from the matrix in equation (23),

$$\widehat{\mathbf{M}} = \begin{bmatrix} 0.833 & -0.388 & -0.165 & -0.317 & -0.330 & -0.161 \\ -0.090 & 1.049 & -0.058 & 0.227 & 0.234 & -0.056 \\ -0.169 & -0.340 & 0.837 & -0.396 & -0.343 & -0.173 \\ -0.058 & 0.222 & -0.094 & 1.039 & 0.224 & -0.057 \\ -0.065 & 0.218 & -0.071 & 0.217 & 1.069 & -0.102 \\ -0.157 & -0.323 & -0.158 & -0.312 & -0.384 & 0.840 \end{bmatrix}. \tag{26}$$

The results of equation (26) clearly show that the resulting model differs significantly from the model obtained in the first calibration. However, the mean absolute error percentage of the second calibration remains at MAEP = 0.35 %. This result validates that the accuracy of calibration results is robust to the placement of Hall sensors because the sensor-network measurement matrix is identified as a complete 6×6 matrix in the proposed calibration method.

IV. Performance Metrics of Magnetic-Flux-Generation Efficiency

Building on the calibrated lumped-parameter model, position-dependent metrics are formulated to quantify magnetic-flux-generation performance. These metrics enable direct and standardized comparisons among alternative actuator designs and realized physical multipole systems.

A. Quantitative performance metrics

Three performance metrics are introduced to quantify the actuator's performance. The actuation gain ${}^{B}\hat{g}_I$ offers an overall capacity measure of the multi-pole actuator. Clearly, the greater the gain is the greater the capacity of the actuator. Two additional position-dependent performance metrics are introduced to yield more advanced quantitative performance measures. The 3×6 position-dependent actuation gain matrix for flux generation is defined as follows,

$${}^{B}\mathbf{G}_I(\mathbf{p}) = \mathbf{S}\left(\mathbf{p}/\hat{\ell},\hat{\mathbf{e}}\right){}^{B}\widehat{\mathbf{H}}_I. \tag{27}$$

By decomposing the gain matrix via singular value decomposition (SVD), ${}^{B}\mathbf{G}_I(\mathbf{p}) = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$, one obtains a set of 3 singular values: ($\sigma_1 \geq \sigma_2 \geq \sigma_3$), the three bases of the 3D linear output vector space ($\mathbf{U} = [\mathbf{u}_1 \;\; \mathbf{u}_2 \;\; \mathbf{u}_3]$), and the six bases of the 6D linear input vector space ($\mathbf{V} = [\mathbf{v}_1 \;\; \cdots \;\; \mathbf{v}_6]$). It is worth noting that the standard notations for SVD are used, therefore, the $\mathbf{V}$ matrix here is not the combined voltage matrix $\mathbf{V}$ defined previously. The model prediction can then be cast as follows,

$$\mathbf{b}(\mathbf{p}) = {}^{B}\mathbf{G}_I(\mathbf{p})\mathbf{I} = \sum_{k=1}^{3}\sigma_k(\mathbf{p})[\mathbf{v}_k^T(\mathbf{p})\mathbf{I}]\,\mathbf{u}_k(\mathbf{p}). \tag{28}$$

This expression uses the concept of linear mapping between two linear vector spaces to abstractly shows the generation of the magnetic flux, wherein the input-current vector $\mathbf{I}$ is projected to each individual right-singular vectors $\mathbf{v}_k^T(\mathbf{p})\mathbf{I}$ in the input vector space, scaled by the respective corresponding singular value $\sigma_k(\mathbf{p})$, and mapped the scaled outputs to each the individual left-singular vectors $\mathbf{u}_k(\mathbf{p})$ in the output vector space.

In the six-input system, the input-current vector constrained by a unit sphere ($\|\mathbf{I}\| \leq 1$) maps to an ellipsoid in the three-dimensional flux space. A performance indicator is defined proportional to the cubic root of the volume of this "flux-generating ellipsoid" to measure the position-dependent efficiency of flux generation,

$$\mathcal{E}\left({}^{B}\mathbf{G}_I(\mathbf{p})\right) = \sqrt[3]{\prod_{k=1}^{3}\sigma_k(\mathbf{p})}. \tag{29}$$

A condition number is calculated using the maximum and minimum singular values to measure how "balanced" or isotropic the actuator design is,

$$\kappa\left({}^{B}\mathbf{G}_I(\mathbf{p})\right) = \frac{\sigma_{min}(\mathbf{p})}{\sigma_{max}(\mathbf{p})} = \frac{\sigma_3(\mathbf{p})}{\sigma_1(\mathbf{p})}. \tag{30}$$

This isotropy index can be used to quantify the magnetic-flux manipulability of multipole electromagnetic actuators.

B. Performance measures of hexapole system design

The proposed quantitative performance metrics—the efficiency and isotropy indices—were used to evaluate the hexapole system design discussed previously.

For simplicity, instead of showing the position-dependent efficiency index within the three-dimensional spherical workspace with the design radius of 500 μm, Figure 9 only shows the spatial distribution of the efficiency index in the x-y and x-z planes (the distribution in the y-z plane is similar). The results

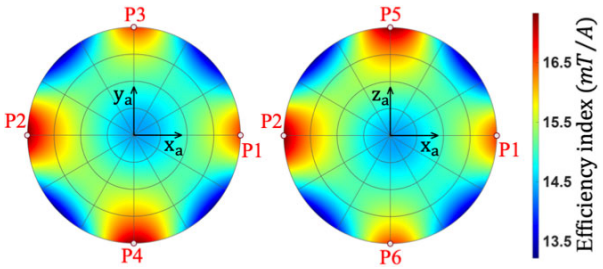


Fig.9. Position-dependent efficiency index within a workspace with a radius of 500 μm: in the x-y plane (left) and the x-z plane (right).

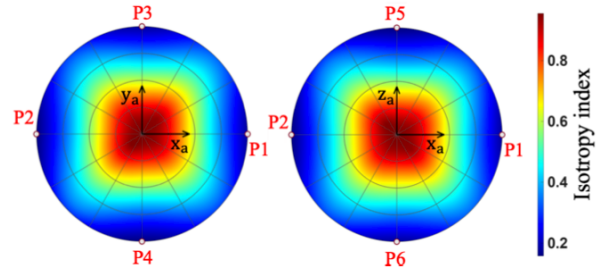


Fig.10. Position-dependent isotropy index within a workspace with a radius of 500 μm: in the x-y plane (left) and the x-z plane (right).

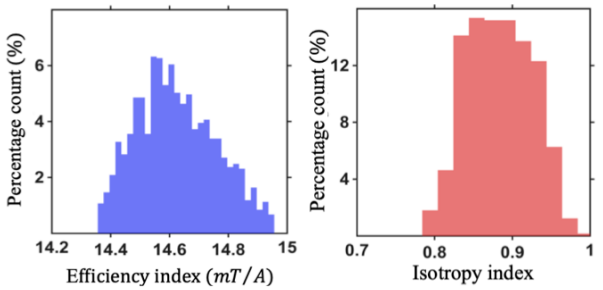


Fig.11. Histogram of the position-dependent quantitative performance metrics within a workspace with a radius of 150 μm: efficiency index (left) and isotropy index (right).

show that although the spatial distribution of the efficiency index is not perfectly symmetrical, its maximum values all occur near the tips of individual poles.

Figure 10 shows the spatial distribution of the isotropy index in the *x-y* and *x-z* planes (the distribution in the *y-z* plane is similar). The results show that while the volume of the flux-generation ellipsoid, and thus the efficiency index, reaches its maximum values for the spatial locations near the tips of individual poles, the shape of the ellipsoid is elongated, leading to a decrease in the isotropy index, and thus a decrease in the magnetic-flux manipulability.

Figure 11 shows histograms of two position-dependent quantitative performance indices within a workspace with a radius of 150 μm. The mean value of the position-dependent efficiency index is 14.625 mT/A, with a standard deviation of 0.1353 mT/A. The mean value of the position-dependent isotropy index is 0.884, with a standard deviation of 0.0413. The results indicate that when the radius of the workspace is limited to 30 % of the design value, the spatial variation of both the efficiency and isotropy indices is sufficiently small to be considered uniformly distributed.

These metrics can be used to enable direct, standardized comparisons across alternative actuator designs during the design stage and establish a unified quantitative foundation for evaluating and benchmarking multipole electromagnetic actuators and magnetic-tweezers systems.

V. CONCLUSION AND DISCUSSION

This work has presented a framework for lumped-parameter modeling, calibration, and quantitative performance evaluation of magnetic-flux generation in multipole electromagnetic actuators and magnetic-tweezers systems. The modeling formulation employs a multiple-charge representation that supports both input-current-based and sensor-voltage-based parameterizations, enabling efficient prediction of magnetic-flux distributions without reliance on computationally intensive FEM analysis.

A five-phase calibration procedure combining nonlinear optimization and analytical calculations was developed to identify a complete lumped-parameter model. For a hexapole configuration, the method successfully estimated 18 geometric parameters, a global actuation gain, and a normalized flux-distribution matrix using only 13 calibration points. Validation against FEM simulations of a hexapole magnetic-tweezers design with integrated Hall sensors demonstrated a mean absolute error percentage of 0.35 % across 1,771 nodes within a 150 μm workspace, confirming both the accuracy of the modeling approach and the effectiveness of the calibration procedure.

Based on the calibrated model, position-dependent metrics were formulated to quantify flux-generation efficiency and magnetic-flux manipulability. These metrics provide standardized, quantitative measures for comparing alternative actuator designs and are well suited for early-stage design evaluation. The resulting lumped-parameter models ~~current~~-based or voltage-based ~~of~~fer a computationally efficient surrogate for FEM simulations and support rapid assessment of flux-generation performance and manipulability in multipole systems.

Although the framework was validated using FEM generated data, the modeling and calibration methodology establishes a mathematical foundation applicable to physical actuators through experimental measurement. In particular, the sensor-voltage-based formulation mitigates uncertainties associated with remanence and hysteresis, making it especially advantageous for calibration and control of realized multipole electromagnetic actuators.

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